

MD Faiyaz Rizwee

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I recently completed my M.Sc. in Bioinformatics and have hands-on experience in web development, database development, and bioinformatics tool development. I am passionate about applying computational approaches to solve biological problems and have a strong interest in machine learning, large language models (LLMs), and retrieval-augmented generation (RAG) systems.

Education

Jamia Millia Islamia M.Sc. Bioinformatics, CGPA: 9.68	New Delhi, India 2024 – 26
B.N. College, Patna University B.Sc. Biotechnology, CGPA: 7.79	Patna, India 2019 – 22
Rajkiyekrit Town inter college Intermediate, Percentage: 73.6	Bihar, India 2017 – 19

Skills

Web Development: HTML5, CSS3, JavaScript (ES6+), React.js, FastAPI.

Database Development: Python, Django, PostgreSQL, REST APIs.

Structural Biology & Drug Discovery: AutoDock, AutoDock Vina, PyRx, Molecular Docking, Protein–Ligand Interaction Analysis.

NGS & Genomics: SRA Toolkit, HISAT2, Bowtie2, SAMtools, GATK, CNVkit, DESeq2, Sequence Alignment, Variant Calling (SNP/INDEL), RNA-Seq Analysis.

Machine Learning & Data Science: Supervised Learning (Classification, Regression), Scikit-learn, Pandas, NumPy, Data Preprocessing, Feature Selection, Model Evaluation, Data Visualization.

Systems Biology & Network Analysis: Cytoscape, CellDesigner, String, Network Construction, Pathway Analysis.

Bioinformatics Tools & Libraries: BLAST, Mercator, ToxinPred, SignalP, TargetP, RDKit, AMPScanner, ESMFold, Blast2GO.

Experience

BRIC - NIPGR (As Intern)

Currently Working from Jan 2026

I developed a comprehensive plant peptide database using Django integrating sequence and functional annotations, and I am also working on a machine learning project for accurate prediction and classification of true open reading frames (ORFs).

Access Database - <https://argue-upriver-encroach.ngrok-free.dev/plantpepdb2/>

Jawaharlal Nehru University (summer Internship)

2 Months

Performed web scraping using Python to collect epilepsy-associated gene data, followed by manual curation and validation of relevant genes. Selected the top 300 genes to construct a biological interaction network using Cytoscape, and conducted network analysis to identify key hub genes as potential biomarkers, strengthening skills in bioinformatics, data processing, and network-based analysis.

Boulevard Tech

Webinar attended

Participated in a hands-on molecular docking workshop using AutoDock Vina and PyRx, where I learned the fundamentals of protein–ligand interactions, docking protocols, and analyzing binding affinities, gaining practical exposure to computational drug discovery techniques.

NextGen Helper

2 Months

Worked on developing and executing NGS data analysis pipelines, including sequence mapping/alignment, SNP and INDEL identification, and variant analysis. Gained hands-on experience with tools such as HISAT2, Bowtie2, SRA Toolkit, SAMtools, and CNVkit, while strengthening skills in omics data processing, workflow design, and interpretation of high-throughput sequencing data.

BRIC – NIPGR

One day symposium

I gained valuable insights into modern genomic data science tools and workflows, including approaches for handling large-scale sequencing data, data integration, and interpretation for biological applications. The symposium also enhanced my understanding of emerging trends such as AI/ML in genomics, translational research, and how computational methods are applied to solve real-world problems in plant and biomedical sciences.

Mind Luster

Certification

Completed a certification in “*Supervised Machine Learning with Python*” from Mind Luster, where I gained practical knowledge of classification and regression algorithms, model training and evaluation, and hands-on experience using Python libraries such as scikit-learn for building predictive models.

Projects

Gene2Therapy

<https://biocontext-gene2therapy.streamlit.app/>

Streamlit-based tool integrating the pyDESeq2 package to perform differential gene expression (DEG) analysis from user-provided expression matrices and generate volcano plots. Implemented features for gene list-based analysis including gene mapping to NCBI annotations, KEGG pathway enrichment, disease association retrieval via OpenTargets, identification of therapeutic compounds for drug discovery, and interactive network visualization. Gained experience in full-stack bioinformatics tool development, data integration from multiple biological databases, and building user-friendly analytical platforms.

Green Valley

<https://school-demo-six-drab.vercel.app/>

Developed a School Management Web Application using Django as part of hands-on full-stack development experience. Implemented features for student and teacher management with database integration using Django ORM, designed a clean user interface using HTML and CSS, and deployed the application on Render. Gained practical experience in backend development, routing, database handling, and deploying web applications in a production environment.

Variant calling

Performed sequence alignment using Bowtie2, followed by variant calling with GATK, SNP/INDEL filtering, and CNV detection using CNVkit. Gained practical experience in variant analysis workflows, genomic data processing, and interpretation of structural and sequence-level variations.

RNA-seq Analysis

Executed a complete RNA-Seq analysis pipeline including quality control (QC), sequence alignment using HISAT2, transcript quantification, and differential gene expression analysis using DESeq2, followed by result visualization. Gained hands-on experience in end-to-end NGS workflow development, data preprocessing, statistical analysis of gene expression, and biological interpretation of results.

Publication

Patient-centric care and digital health tools

<https://doi.org/10.1016/bs.pbr.2025.08.003>

Ali, A. M., Shah, S. N., Shamim, T., Khan, A., **Rizwee, F.**, et al. (2025). Patient-centric care and digital health tools. *Advances in Protein Chemistry and Structural Biology*, Elsevier.